

Chapter 18 — Fundamentals of Laparoscopic and Robotic Urologic Surgery — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Working pressure 10–12 mm Hg**; never >20 mm Hg for long.
- **Never insufflate** until aspiration, saline, drop and advancement tests all pass.
- **Open (Hasson) access** when adhesions expected; **Palmer point** for closed entry.
- **Blunt trocars**: about 5× less epigastric injury; hernia 1.8% → 0.19%.
- **Gas embolism**: end-tidal CO₂ ↑ then ↓, saturation ↓ → desufflate, left lateral, head down, 100% O₂.
- **Thermal bowel injury** may present up to **18 days** later; CT with oral contrast.
- **Monopolar bowel burn** → resect with 6-cm margins; bipolar → excise visible area.
- **Exit**: inspect every port at 5 mm Hg; remove under vision; scope last.
- **Never metal trocar + plastic retainer** (capacitive coupling).
- **No shoulder braces in Trendelenburg** (brachial plexus).

PREOPERATIVE PREPARATION

Contraindications:

- **Uncorrectable coagulopathy** • **untreated obstruction** • **abdominal wall infection** • **massive haemoperitoneum** • **generalised peritonitis** • **malignant ascites**.

Special cases:

- **COPD** (chronic obstructive pulmonary disease) → arterial gases + lung function tests pre-op.
- **Obesity**: renal or adrenal laparoscopy still beats open; prostatectomy and cystectomy riskier (BMI >30).
- **Pregnancy**: 8–12 mm Hg; second trimester; Palmer point; difficult after 20 weeks.
- **Hip prosthesis cement** → pelvic fibrosis.
- **Aneurysm** → left upper quadrant Veress or Hasson.
- **Robot >20 cases** → better outcomes.

Preparation:

- **No mechanical bowel prep** (even cystectomy with conduit).
- **ERAS** (enhanced recovery after surgery, 2001): solids 6 h, carbohydrate drink 2 h; stay ~30%, complications ~50%.
- **Group and screen**: nephrectomy (transfusion 3–12%), prostatectomy (2.5%).
- **Crossmatch**: partial nephrectomy, cystectomy, caval thrombus (early experience).

OPERATING ROOM AND TEAM

- **Lateral**: lower leg 45°, upper straight; axillary roll 3–4 fingers below axilla.
- **Rhabdomyolysis risk**: male, BMI ≥25, kidney rest, full flexion; rest <20–30 min.
- **Plain mattress** beats gel or egg-crate for flank pressure.
- **Heparin 5,000 U SC** optional; compression stockings for all.
- **Xi**: 8-mm camera any port, boom, laser targeting; **Si**: 12-mm camera.
- **SP** (single port): single arm, articulating camera, 360° access.
- **Retroperitoneal**: true 90° lateral; team faces back; robot over head.
- **Pelvis**: 30° Trendelenburg; Si robot between legs; Xi either side, no stirrups.

PERFORMING THE PROCEDURE

Veress:

- **14 G**, 2 mm OD (outer diameter); flow check 2 L/min → ≤2 mm Hg.
- **Two pops**: fascia, peritoneum.
- **Umbilicus**: non-obese angle to pelvis; obese perpendicular.
- **Lateral**: 2 fingers medial + 2 superior to ASIS (anterior superior iliac spine).

- **Advancement test**: 1 cm free.
- **Insufflate** 9 L/min to 15 mm Hg; initial pressure <8–10 = free.

Open access and retroperitoneum:

- **Hasson**: 2-cm infra-umbilical incision; two 0 silk stays.
- **Hand port** 6.5–7.5 cm; incision (cm) = glove size.
- **Retroperitoneal**: 2–2.5 cm below 12th rib tip, midaxillary line; psoas behind, kidney in front.
- **Balloon** max 800 mL (40 pumps); oblong retroperitoneal, round pelvic.
- **Gaur dilator** (1992): glove on 8-Fr catheter, 110 mm Hg.
- **Extraperitoneal**: 1.5–2 cm infra-umbilical; finger to pubis.
- **Balloon port** → less emphysema and hypercarbia retroperitoneally.

Trocars:

- **Primary trocar**: perpendicular (upper abdomen) vs 70° caudad (pelvis).
- **Blunt vs bladed**: bleeding 3% vs 9%; complications 3% vs 10%.
- **AirSeal**: stable pneumoperitoneum at 8–12 mm Hg; smoke evacuation.
- **Suturing angles**: <55° to horizontal; 25–45° between instruments.
- **Robot ports** 8–10 cm apart; assistant port caudal; black line below wall.

Instruments:

- **Thermal spread**: monopolar 5–7 mm; bipolar 1–6; LigaSure 1–3; ultrasonic ≈1.
- **LigaSure** seals ≤7 mm; one application only.
- **Ultrasonic** 25–55 kHz; pacemakers OK; blade >200 °C, cools up to 45 s.
- **Staples**: 2.0 grey / 2.5 white = vascular; 3.8 blue / 4.8 green = thick.
- **Endo-TA** on donor renal vein → longer vein.
- **Never staple across clips**; no diathermy near clips.
- **Lapra-Ty**: 3.5 mm, polydioxanone, 14 days; avoid with 4-0.
- **Hem-o-lok** donor artery: ≥2 clips, 2-mm cuff; donor use contraindicated by maker.
- **LapSac**: most tear-resistant, 2 kg, morcellation; ring forceps.
- **EndoWrist**: 7 degrees of freedom, 90° articulation, 10-use limit.
- **5-mm robotic scope** = 2D.
- **Robot failure** <1%; malfunction 0.4%.

Exit:

- **Bladed**: close all 10–12-mm sites (children ≥5 mm).
- **Blunt**: close midline >10 mm only.
- **Hand port** removed and closed first.
- **Carter-Thomason** for BMI >30.

PHYSIOLOGY

- **CO₂** standard; **helium** for COPD or hypercarbia (embolism risk; start with CO₂).
- **N₂O** supports combustion → no diathermy; air and O₂ abandoned.
- **12 mm Hg** = no stroke-volume change; **10 mm Hg** → less oliguria.
- **40 mm Hg** → venous return, output and heart rate ↓.
- **Oliguria** = renal vein and parenchymal compression; mannitol 12.5–25 g, dopamine 2 µg/kg/min; target UO (urine output) >100 mL/h.
- **CVP** (central venous pressure) **unreliable** under pneumoperitoneum.
- **COPD**: arterial gases if insufflation >1 h; continue in recovery.
- **CO₂ absorption plateaus** after 30–60 min.
- **Head-down**: CO ↑ (cardiac output), MAP ↑ (mean arterial pressure), HR ↓ (heart rate); head-up: CO ↓.
- **Hypercarbia peaks after extubation.**

COMPLICATIONS

- **Contemporary complications 13–22%**; mortality 0.03–0.08% historically.
- **Commonest**: vascular, then adjacent organ.
- **Training**: course + preceptorship (≥5 cases) + mentor.
- **Preperitoneal**: pressure spike at about 500 mL; camera sees fat.
- **Bowel insufflation**: asymmetry, flatus, high pressure <2 L.
- **Barotrauma** >15 mm Hg → restart at 10.
- **Pneumothorax 1.6–4%**: 16 G, 2nd–3rd intercostal space, MCL (midclavicular line) → drain.
- **Pneumopericardium** about 0.8%; trocars below 12th rib.
- **Vascular injury 0.11–2%**: aorta and common iliac artery commonest; retroperitoneal > pelvic.
- **Trocar in vessel** → leave in, laparotomy along trocar.
- **Venous bleed** → pressure 25 mm Hg, tamponade; pinch IVC (inferior vena cava) walls.
- **Port-site bleed**: cantilever to find; Keith needle over bolster.
- **Bladder injury**: indigo carmine; always repair.
- **Free air** persists up to 9 days; left shift > white count.
- **Nerve injury about 2.8%**; brachial plexus commonest.

- **Hand-port hernia 4–7%** → interrupted non-absorbable closure.
- **Transverse incision**: hernia 5.4% → 0.4% after robotic prostatectomy.
- **DVT prophylaxis** (deep vein thrombosis): compression 48–72 h; high risk + heparin 5,000 U q12h.
- **Hand-port wound infection** about 9%.
- **Rhabdomyolysis**: CK (creatine kinase) >5,000; kidney rest <1 h; cases <5 h.
- **Bowel entrapment**: day 2–3, ileus + port tenderness.

TRAINING

- **E-BLUS** (EAU Basics in Laparoscopy, 2013): peg transfer, circle cutting, needle guidance, suturing.
- **FRS** (Fundamentals of Robotic Surgery): RCT → better speed and accuracy.
- **VR** (virtual reality) beats box or none for novices; **swine** commonest animal.
- **Practice suture**: 3-0 on SH (small half-circle) needle.
- **Mini-residency** (5 days): about 90% doing robotic prostatectomy at 3 years.

CLASSIC TRAPS

- **Trap**: lift fascia during Veress — **Actually**: widens preperitoneal gap only.
- **Trap**: Veress angled to pelvis in obese — **Actually**: perpendicular (umbilicus lower).
- **Trap**: shoulder braces in steep Trendelenburg — **Actually**: brachial plexus injury; use straps or beanbag.
- **Trap**: trust CVP intraoperatively — **Actually**: falsely raised.
- **Trap**: fluid boluses for oliguria — **Actually**: lower pressure; mannitol or furosemide or dopamine.
- **Trap**: ultrasonic = safe near bowel — **Actually**: blade >200 °C after firing.
- **Trap**: repeat LigaSure for security — **Actually**: weakens the seal.
- **Trap**: catheter drainage alone for trocar bladder injury — **Actually**: always repair.
- **Trap**: remove trocar from injured vessel — **Actually**: leave it in, open along it.
- **Trap**: normal day-1 course excludes bowel injury — **Actually**: thermal necrosis up to 18 days.